Wei (Tzu-Wei) Lee

Statistics/ML Model Evaluation | Complex Dynamical Systems
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EDUCATION

Vanderbilt University, Nashville, USA **Aug. 2021 – Present**
Ph.D. Candidate in Bioengineering and Biomedical Engineering | Expected Graduation: 2026
Taipei Medical University, Taipei, Taiwan Sept. 2016 – June 2020
Bachelor of Science in Biomedical Engineering – Last 60 GPA: 3.9/4.0
Visiting Scholar @ University of Arizona Jan. 2019 – Feb. 2019

SKILLS

Languages: Python (JAX/XLA, PyTorch), C++, C#, Go, MATLAB, SQL, R, Julia, Node.js, HTML
AI Infra & MLOps: vLLM, Triton, CUDA, SGLang, Kubernetes, Linux, LongChain
Cloud Architecture & DevOps: AWS, Docker, Kubernetes, Ray, Linux, GCP.

RESEARCH EXPERIENCE

Graduate Research Assistant @ Vanderbilt University Imaging Institute *Aug. 2021 – Present*

- **Architected a multi-modal data ingestion and representation pipeline for large-scale biological datasets** (HCP, UK Biobank), achieving high-dimensional feature isolation for disease signatures by integrating fMRI and genomics data through a Spatiotemporal Masked Autoencoder, with findings selected for an upcoming conference presentation (June 2026).
- **Designed a novel frequency-domain evaluation framework** for neural time-series data, combining spectral analysis and dynamical system modeling to identify region-specific failure modes. (Manuscripted paper)
- **Delivered the first end-to-end mouse spinal cord fMRI pipeline** by designing a custom hardware-software interface involving 3D-printed prototypes, C++ pulse sequences, and statistical analysis, enabling new research capabilities in spinal imaging. (Manuscripted paper)
- **Engineered a monkey palm motion tracking system** using deep learning, implementing iterative preprocessing improvements that reduced error rates across diverse experimental conditions.

WORK EXPERIENCE

Research Assistant @ Best-Image & Siemens Health, Taiwan Sept. 2020 – Jan 2021

- **Developed Python-based ML evaluation pipelines** for neuroimaging (including MRI denoising and segmentation), implementing state-of-the-art processing to enhance quality for clinical diagnostics.
- Developed practical IDEA pulse sequence programming in ASL imaging sequence on VE11C in 3T Siemens PRISMA MRI scanner

Intern @ Engineering & Technology Department, Dynamic Medical Technologies Inc. July – Sept. 2019

- Executed installation, maintenance, and troubleshooting of plastic surgery laser instruments, ensuring optimal performance and reliability.

Research Assistant @ Applied Mechanics, National Taiwan University. May 2017 – July 2018

- Designed & prototyped microchannel chips to optimize blood centrifugation without electricity.
- Conducted ELISA protein analysis, developed 3D models, & fabricated a microchannel chip.
- Gained hands-on semiconductor fabrication experience, contributing to micro-fluidic chip.

SELECTED Applied AI & Infrastructure Projects

High-Performance Speculative Decoding LLM Inference Optimization | *Lead Developer*

- Optimized LLM inference latency by 1.20x for large-scale deployments by architecting a JAX/XLA pipeline featuring 2-GPU sharding and speculative decoding, validated through rigorous profiling of 50+ vLLM configurations.

AWS-Specific Project (Nova, SFT & SageMaker)

- Engineered a multi-agent orchestration framework using Amazon Nova models, achieving standardized "solution-format" outputs from raw homework data by implementing SageMaker-based Supervised Fine-Tuning (SFT) on S3-hosted datasets.

SELECTED PUBLICATIONS

- Kevin D. Harkins, **Tzu-Wei Lee**, Jonathan B. Martin, Hannah E. Alderson, John C. Gore and Mark D. Does. Diffusion MRI with a Multi-Shot Rosette Readout SSRN 2026 (Under Review)
- Yikang Li, Lyuan Xu, **Tzu-Wei.Lee**, Kurt G. Schilling, Muwei Li, Zhongliang Zu, Zhaohua Ding, Yurui Gao, John C. Gore. "Revealing White-Matter Functional Fingerprints with Graph Representation Learning". OHBM, Bordeaux, France, 2026 (Poster Presentation).
- **Tzu-Wei Lee**, Feng Wang, Yikang Li, Li Min Chen, John Gore. Mapping Vascular Reactivity in Mouse Spinal Cord at 15.2T ISMRM 2026 (Oral)
- **Tzu-Wei Lee.**, Feng Wang, Anirban Sengupta, Arabinda Mishra, Limin Chen, and John C. Gore. Resting-state functional connectivity in mouse cervical spinal cord at 15.2T. ISMRM 2025
- **Tzu-Wei Lee**, John C. Gore & Kevin D. Harkins. Diffusion MRI of Murine Spinal Cord with Multi-Shot Rosette Readouts at 15.2T. ISMRM Diffusion Workshop 2025
- **Tzu-Wei Lee,** Chen, C. Y., Chen, K., Tso, C. W., Lin, H. H., Lai, Y. L. L., ... & Liu, H. S. (2021). Evaluation of the swallow-tail sign and correlations of neuromelanin signal with susceptibility and relaxations. Tomography, 7(2), 107-119.
- Lai, Y. L. L., Chen, K., **Tzu-Wei Lee**, Tso, C. W., Lin, H. H., Kuo, L. W., ... & Liu, H. S. (2021). The Effect of the APOE-ε4 Allele on the Cholinergic Circuitry for Subjects with Different Levels of Cognitive Impairment. *Frontiers in Neurology*, *12*, 651388.
- **Tzu-Wei Lee**, Tso, C.-W., Chen, K., Chen, C.-Y., & Liu, H.-S. Visualization of Nigrosome-1 with improved contrast-to-noise ratio and correlation with signals on neuromelanin-sensitive MRI. ISMRM 2020
- Chen, K., **Tzu-Wei Lee**, Tso, C.-W., Lai, L. Y.-L., Lin, H.-H., Hsu, F.-T., & Liu, H.-S. Radiogenomic analysis using dynamic histogram parameters of MR perfusion-weighted imaging in glioblastoma. ISMRM 2020
- Liu, H. S., **Tzu-Wei Lee**, Tso, C.-W. and Chen, K. A Practical Way to Improve Contrast to noise Ratio (CNR) in Visualizing Nigrosome 1, 5th International Workshop on MRI Phase Contrast & Quantitative Susceptibility Mapping. Sept 2019. Seoul, Korea

TEACHING EXPERIENCE

- MATLAB Programming & Application, Fall 2020.
- Medical Imaging & Neuroscience, Fall 2020 & 2019.
- Medical Imaging Systems, Fall 2019
- Neuroscience & Molecular Imaging, Fall 2019
- Advanced Medical Imaging Approach, Spring 2019